

ABB Ltd. (ABBN.S): Takeaways from webcast on new 800VDC product portfolio

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Yesterday (24th September), we attended ABB's webcast on its new Infinitus source-to-rack Direct Current portfolio for AI datacenters. Here are our key takeaways:

- **ABB expects 25-40% of new datacenter additions in 2030 to be based on Direct Current with total capacity installed to increase by 170GW in the next five years (from c.80GW at the end of 2025). This compares with the 21%/17% expectations for US/Europe quoted in our recent expert call (see [here](#)).** As power demand increases from 30kW per rack a few years ago to potentially 1MW per rack in the near future, 800VDC will need to be implemented to reduce power conversion losses and copper requirements. ABB demonstrates that for a 500MW data center, replacing traditional AC technology with DC technology could reduce conversion losses by >5%, representing an additional 25MW of power and c.\$100-300mn additional revenues. Furthermore, DC technology could reduce copper requirements for a 500MW DC by c.10,000 tonnes representing c.\$150mn in cost savings. ABB sees DC solutions being implemented not only in AI training but also inference due to the architecture's efficiency gains. However, our expert believed that the use of 800VDC for AI inference is highly unlikely given it requires maximum 150kW per rack.
- **Capital investment required for electrical equipment should remain broadly similar but white space is expected to decrease from 45% to 25% of total power infrastructure.** Whilst the total capital expenditure for a data center will decrease (due to reduced copper requirements), ABB believes investments required for electrical equipment will be broadly stable (+/-10% depending on the exact architecture). However, ABB expects this capital to shift from the white space to the grey space with white space potentially decreasing from 45% to 25% of the total power infrastructure. Given ABB has a more significant proportion of revenues from the grey space, management expects its \$2mn content per MW for AC architecture to increase as datacenters move to direct

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current. **This compares with our expert call that sees the whole powertrain requiring \$4mn per MW for AI training and \$2.6mn per MW for AI inference.**

- **Infinitus is the industry's first source-to-rack DC product portfolio with parts already available and the full solution expected to be available within 12m.** The portfolio consists of five building blocks including the powertrain, AC to DC conversion, DC power distribution, DC power protection and cooling optimization. The powertrain delivers stable power from the grid using MV switchgears and the HiPerGuard MV UPS. AC to DC conversion will occur outside the white space for greater power efficiency and smaller footprint using the Infinitus Solid State Transformer and TRUs. DC power distribution and protection is then enabled by DC switchgears, circuit breakers and switchboards. Finally, while ABB does not provide cooling technologies itself, it is a market leader in variable speed drives and high efficiency motors for control of fans and CDUs and also offers batteries for CDUs.
- **ABB believes MV UPSs will not only facilitate the transition to hybrid architectures but will also be necessary for full 800VDC solutions.** ABB's HiPerGuard MV UPS connects directly to 34.5kV grid power, reducing footprint by 30% versus traditional architecture, and is designed in a modular way (customers can connect 20 MV UPS units together to provide a 50MW UPS system block). ABB believes MV UPSs will still be needed in full DC architectures since, whilst there will be alternatives such as BESS, MV UPSs provide increased energy efficiency. Management noted that more players have entered the space which aligns with our expert call saying that Vertiv has also been pitching a MV UPS. ABB sees this as beneficial as it will encourage adoption by hyperscalers which often require double/multi sourcing of electrical equipment. ABB has taken orders for its MV UPS (in the GW range) as well as already installed some.
- **The Infinitus Solid State Transformer provides 3MW+ power with 98.5% efficiency and should be commercially available by 2H27.** ABB sees its SST as a leading product given its design is focused for datacenters, which contrasts with many products on the market that ABB says have been developed for EV charging so do not have sufficiently high power ratings. That being said, our expert call highlighted that Delta Electronics and Heron Power (start-up) are targeting 5MW designs (deemed the sweet spot in terms of size and flexibility for datacenter SSTs) while Eaton will be sampling a 2MW SST with customers by the end of 2026. ABB's Infinitus SST will be first deployed as proof of concept installations (not yet possible to test on real conditions since 1MW racks do not yet exist).
- **ABB has also developed the first IEC-certified solid state breaker which our expert call also highlighted was crucial for the transition to 800VDC given the product is 100x faster than traditional breakers.** Orders for these circuit breakers have already been taken for other types of industries and ABB is currently in discussion with hyperscalers and a few colocators for their application in data centers. Two new frames will be introduced by the end of next year (200A and 500A).

Valuation and Risks

We are Neutral rated. Our 12-month price target is at SFr82. The target price remains based 75% on our EV/IC to ROIC/WACC methodology and 25% on our SOTP, using 6m27/6m28 estimates.

Key upside/downside risks include: (1) A stronger/weaker-than-expected capex environment in key end markets; (2) Market share gains/losses as a result of better/weaker execution; (3) Stronger/weaker-than-expected pricing; (4) Favorable/unfavorable FX volatility; and (5) Value-accretive/dilutive M&A and/or portfolio pruning.

ABBN.S	12m Price Target: SFr82.00	Price: SFr80.50	Upside: 1.9%
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Neutral		GS Forecast			
Market cap: SFr173.2bn / \$208.9bn Enterprise value: SFr177.9bn / \$214.7bn 3m ADTV: SFr183.0mn / \$225.6mn Switzerland Europe Multi-Industry M&A Rank: 3 Leases incl. in net debt & EV?: Yes	Revenue (\$ mn)	12/25	12/26E	12/27E	12/28E
	EBIT (\$ mn)	33,220.0	38,265.5	45,618.2	55,014.8
	EPS (\$)	6,314.0	8,063.1	9,369.1	11,526.7
	P/E (X)	2.64	3.49	3.89	4.99
	EV/EBITDA (ex lease,X)	23.6	27.8	25.0	19.5
	Dividend yield (%)	16.8	20.1	17.4	13.6
	FCF yield (%)	1.5	1.0	1.0	1.0
	CROCI (%)	3.6	2.3	2.8	3.9
	N debt/EBITDA (ex lease,X)	17.2	17.2	20.3	21.3
		0.1	0.1	0.2	0.3
	EPS (\$)	6/26	9/26E	12/26E	3/27E
		0.84	0.88	0.88	0.89

Source: Company data, Goldman Sachs Research estimates, FactSet. Price as of 24 Sep 2026 close.

Disclosure Appendix

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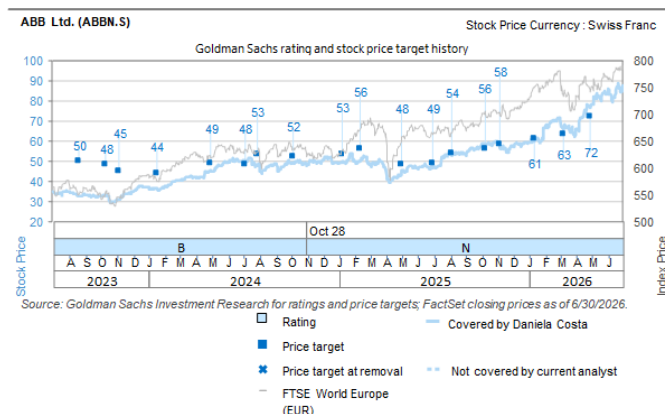
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ABB Ltd. (ABBN.S)

Date of report Target price (SFr) Closing price (SFr)

22-Jul-26	82.00	80.70
07-Jul-26	74.00	83.36
29-Apr-26	72.00	76.68
09-Mar-26	63.00	65.28
11-Jan-26	61.00	60.16
06-Nov-25	58.00	56.58
08-Oct-25	56.00	58.94
06-Aug-25	54.00	52.48
01-Jul-25	49.00	46.69
02-May-25	48.00	45.19
10-Feb-25	56.00	50.08
07-Jan-25	53.00	49.48
04-Oct-24	52.00	49.15
29-Jul-24	53.00	47.77
04-Jul-24	48.00	50.52
30-Apr-24	49.00	44.89
17-Jan-24	44.00	36.10
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11-Oct-23	48.00	33.38

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